

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

The length, y , of a white whale was **162 centimeters (cm)** when it was born and increased an average of **4.8 cm** per month for the first **12** months after it was born. Which equation best represents this situation, where x is the number of months after the whale was born and y is the length, in **cm**, of the whale?

Answer

- A. $y = 162x$
- B. $y = 162x + 162$
- C. $y = 4.8x + 4.8$
- D. $y = 4.8x + 162$

Correct Answer: D

Rationale

Choice D is correct. It's given that the length of the whale was **162 cm** when it was born and that its length increased an average of **4.8 cm** per month for the first **12** months after it was born. Since x represents the number of months after the whale was born, the total increase in the whale's length, in **cm**, is **4.8** times x , or **4.8x**. The length of the whale y , in **cm**, can be found by adding the whale's length at birth, **162 cm**, to the total increase in length, **4.8x cm**. Therefore, the equation that best represents this situation is $y = 4.8x + 162$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.